Question 1: Who are the most valuable customer groups?

Cluster 1 Higher-Value Repeat Customers is the strongest group for a retention campaign.

Measure	Cluster 0	Cluster 1
Customers	93,099	2,997
Mean frequency	1.00	2.12
Mean monetary value	137.63	259.87
Share of customers	96.88%	3.12%

Cluster 1 has higher average purchase frequency and monetary value. Management could test loyalty rewards or personalized offers for this group and measure whether they increase repeat purchases.

Question 2: Which products and categories generate the strongest business performance?

The five leading categories by total item sales are:

Category	Total item sales
Health and beauty	1,258,681.34
Watches and gifts	1,205,005.68
Bed, bath and table	1,036,988.68
Sports and leisure	988,048.97
Computers and accessories	911,954.32

These are sales totals, not profit. Profitability cannot be confirmed without cost and margin information. The notebook results also do not establish individual product rankings.

Question 3: Which factors are linked with poor customer satisfaction?

Delivery timing has a strong observed association with review scores.

- Late deliveries: mean review score 2.57; median 2.
- On-time or early deliveries: mean review score 4.29; median 5.
- Late-delivery share: approximately 8.11%.

Management should investigate delivery delays, estimated delivery accuracy, and logistics performance. However, this analysis shows an association and does not prove that delivery timing is the only cause of poor reviews.

Question 4: Can the company predict a useful business outcome before it happens?

The tuned Random Forest predicts whether a delivered order will receive a low review, defined as a score of 1 or 2. The prediction point is after delivery but before the customer submits a review.

Test metric	Result
Accuracy	89.84%
Precision for low reviews	74.16%
Recall for low reviews	31.41%
F1 score	0.4413
ROC AUC	0.7636

The model is a preliminary candidate, not ready for unattended deployment. Its low recall means it misses many low-review cases, and its perfect training metrics versus weaker test metrics indicate substantial overfitting.

Question 5: How are monthly sales changing, and what should the company expect next?

Monthly item sales reached approximately 1.01 million in November 2017 and remained comparatively high through much of 2018. However, recorded sales then fell to 145 in September 2018 and zero in October 2018.

The forecast's test MAE was approximately 674,829 and RMSE approximately 784,015. It also produced negative future sales forecasts.

These results are not dependable enough for inventory or staffing decisions. The unusually low final months must be investigated before a new forecast is trusted.

Question 6: How can management use the final model and dashboard to make decisions?

The Phase A findings support four practical actions:

1. Test retention campaigns for the higher-value repeat-customer segment.
2. Investigate late deliveries because they are associated with lower review scores.
3. Review the leading sales categories alongside their costs and profit margins.
4. Use the low-review model only as a potential prioritization aid after further validation, with human review.